

ISY 5001: Intelligent Reasoning Systems

Football IQ : Predictive Insights for Winning Strategies

Football match score prediction and data driven insights for team Managers

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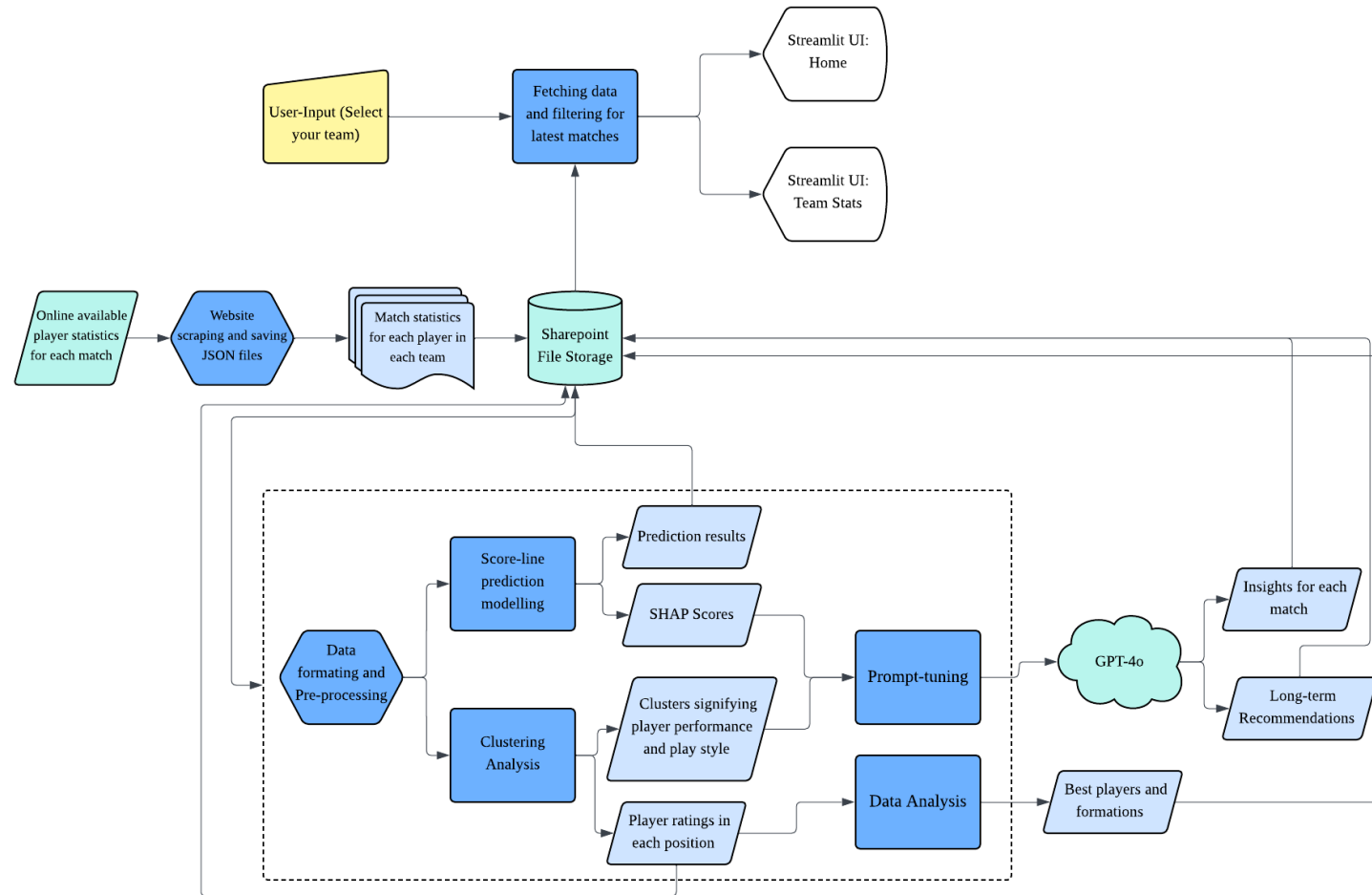
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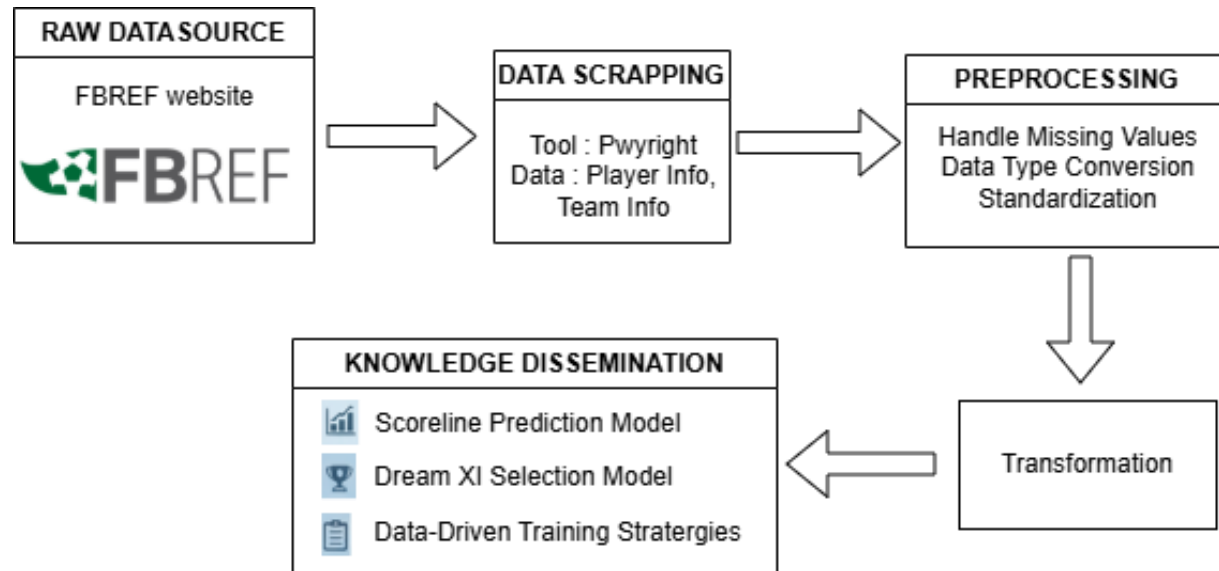
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System Design

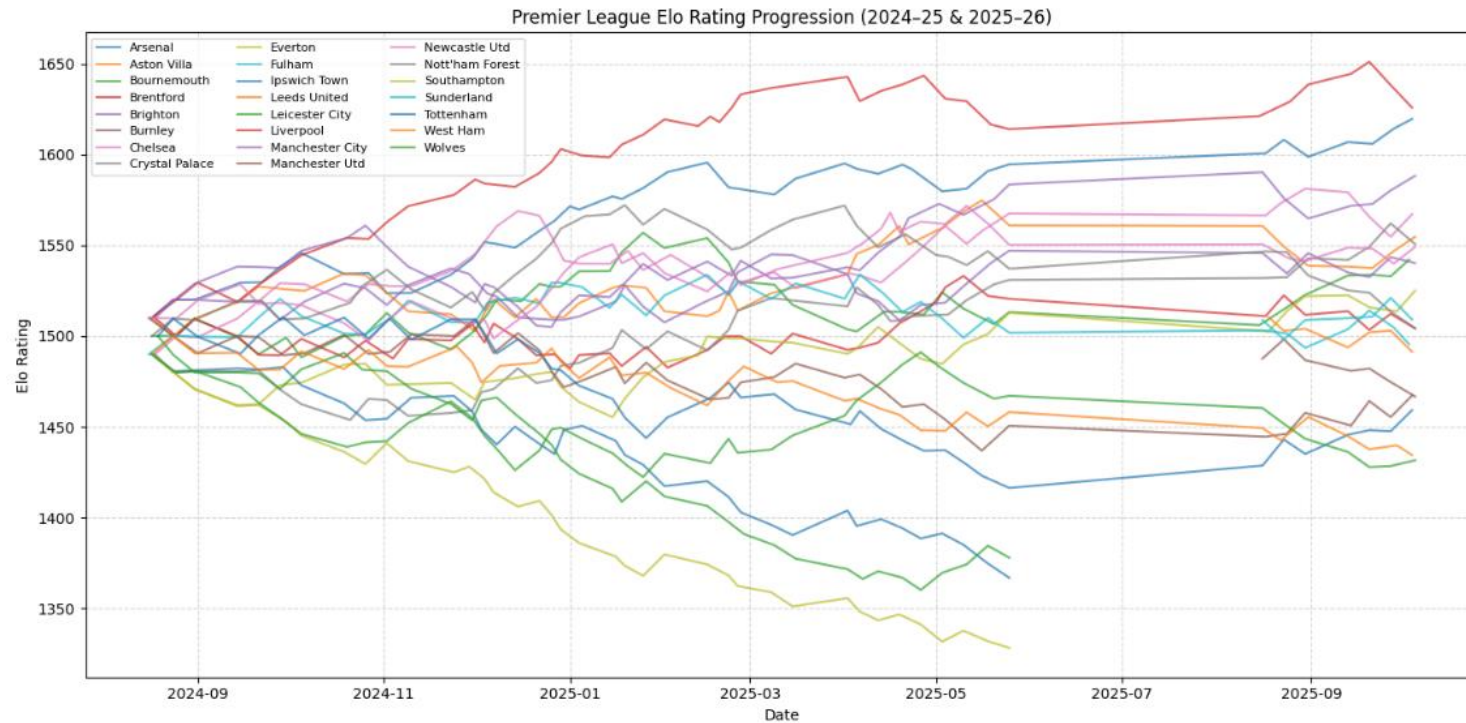


Data Ingestion using Scrapping



- **Raw Data Source** : FBref — detailed Premier League player & team statistic
- **Data Scrapping** : Automated extraction using Pywright for structured football data
- **Preprocessing** : Cleaning, handling missing values, and standardizing formats
- **Feature transformation** : Engineering meaningful performance features for analysis
- **Knowledge & Insights** : Scoreline prediction, Best XI recommendation, Player improvement suggestions

Scoreline Prediction Using Elo Score with XGB



- Elo rating system is a widely used statistical model designed to quantify the relative strength of teams or players based on past performance.
- Elo dynamically adjusts team strength after every match, reflecting performance trends better than static standings.
- Update Rule : $\text{New Rating} = \text{Old Rating} + K(\text{Actual Result} - \text{Expected Result})$

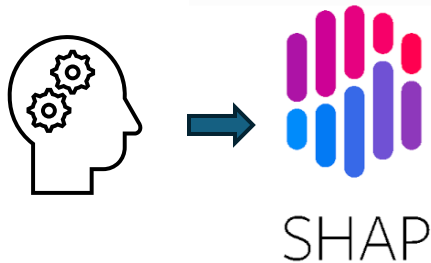
Modelling techniques for Scoreline prediction

Features	Model Used	MAE	MSE
Elo Rating of the team	XGBoost	1.67	2.05
Elo Rating of the team + Embedding of home team based on players position strength + Embedding of away team based on players position strength + Team Strength + Recent Form	XGBoost	0.94	1.4
Elo Rating of the team + Elo Rating of players based on the position played for home and team + Team Strength + Recent Form	XGBoost	0.3	0.2
Selected Features of all players in a team including substitutes for each match	Customer Dense NN architecture	-	0.22
Player position coordinates, positional familiarity and selected features for all player averaged over entire dataset of matches	Linear Regression	1.56	3.96
Player position coordinates, positional familiarity and selected features for all player averaged over entire dataset of matches	MLP Shallow (single layer)	1.44	3.21
Player position coordinates, positional familiarity and selected features for all player averaged over entire dataset of matches	XGBoost	1.58	0.97

Using SHAP to generate insights

SHAP (Shapely Additive)

features are calculated to measure the impact of each feature on the predicted scoreline



```
"Erling Haaland": {  
  "position": "FW",  
  "PrgDist": -0.12080592006444935,  
  "TotDist": -0.01770718604326238,  
  "Crs": -0.015675976872444042,  
  "Tk1W": -0.00666282534599294,  
  "Gls": -0.004588264822959931,  
  "familiarity": -0.003547988533973756,  
  "Tk1+Int": -0.0026135353744030088
```

OpenAI
+
Prompt Tuning

The screenshot shows a match insights dashboard for Arsenal vs Burnley on Saturday, Nov 01, 2025. The predicted scoreline is 1-2. The dashboard includes a section for 'Match Insights & Recommendations' with 'Strategic Recommendations' for three players: Jaidon Anthony (LM), Hjalmar Ekdal (CB), and Loum Tchaouna (RM). Each recommendation includes an 'Action' and 'Reasoning'.

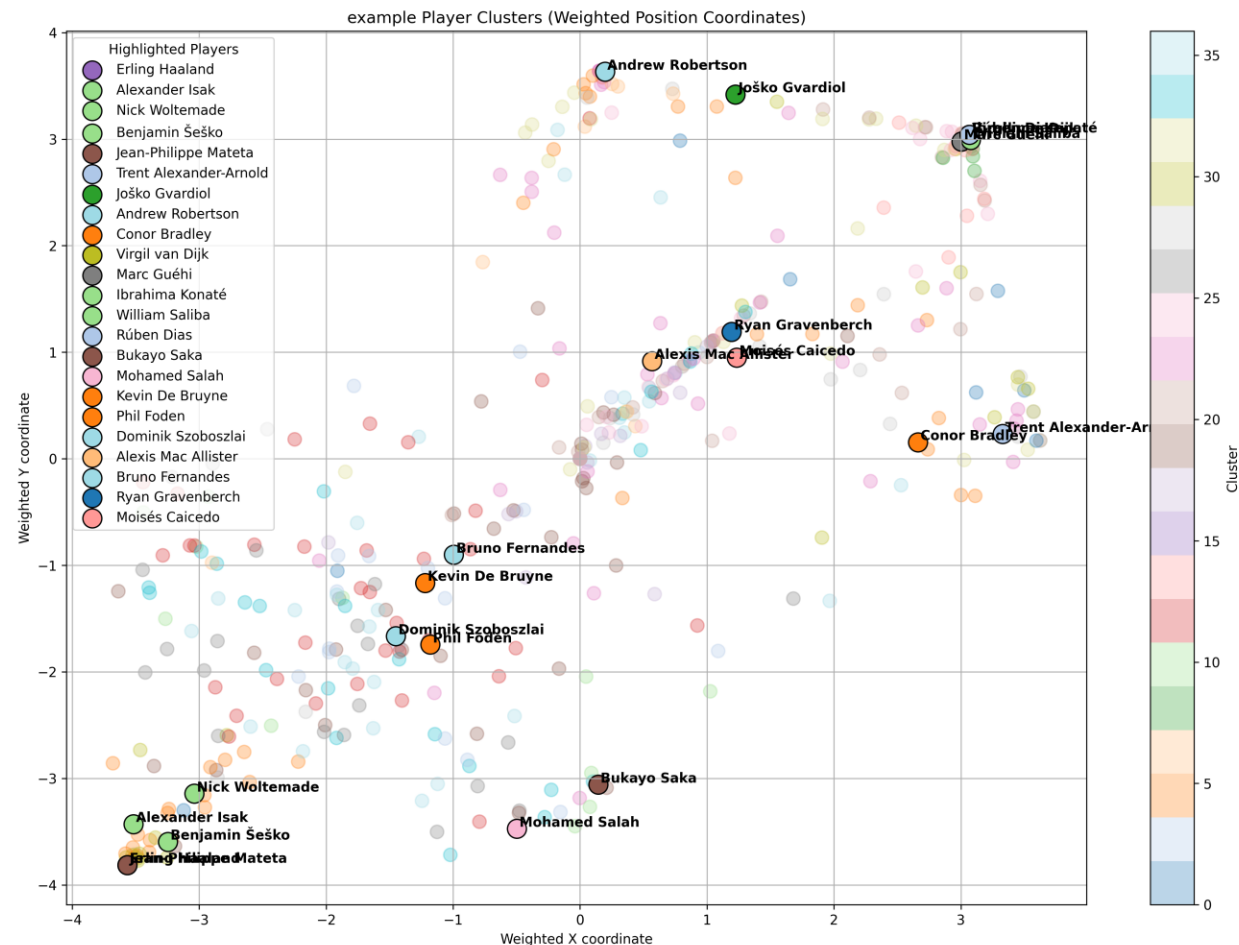
- 1. Enhance Positional Familiarity and Tactical Awareness for Jaidon Anthony (LM):**
 - Action:** Implement targeted training sessions focusing on improving Jaidon Anthony's familiarity with the team's tactical setup and his role within it. This could include match scenarios to enhance his understanding and decision-making.
 - Reasoning:** Jaidon's negative impact from low familiarity suggests he may not be fully integrated into the team's tactical framework, which can hinder his effectiveness. Improving his tactical awareness will help him make better decisions on the field, increasing his ability to support offensive plays.
- 2. Improve Ball Distribution and Progression for Hjalmar Ekdal (CB):**
 - Action:** Conduct passing and ball progression drills aimed at enhancing Hjalmar Ekdal's ability to initiate attacks from the back. Focus on improving his ground game and executing key passes.
 - Reasoning:** As a center-back, Ekdal's negative impact from low Goal-Creating Actions (GCA) and Key Passes (KP) indicates a need to better contribute to the team's attack. Improving his ball distribution will help transition the ball more effectively from defense to attack, increasing the team's goal-scoring probability.
- 3. Boost Dribbling and Ball Retention Skills for Loum Tchaouna (RM):**
 - Action:** Implement dribbling and ball retention exercises to enhance Loum Tchaouna's ability to maintain possession and progress the ball. Focus on one-on-one situations and maintaining possession under pressure to improve his success rate in duels and overall ball control.
 - Reasoning:** Loum's negative impact from low Success Percentage (Succ%) and Won Percentage (Won%) suggests difficulties in retaining possession and winning the ball. Improving his dribbling and ball retention skills will help him maintain possession and create more opportunities for the team to advance the ball and create goal-scoring opportunities from the right flank.

Best performing model
used to predict scorelines
of all upcoming matches,
which in our case is
Xgboost regressor

These are sample SHAP values
for each variable. **Negative
values indicate negative
impact on the score of the
team.**

Recommendation of players insights using Clustering

- Identified players with similar styles through clustering
- Rated players' seasonal performance based on cluster centroids and found their weakness



name	pred_rating	weakness
Ben White	5.4046547	['SoTA', 'T
Bukayo Sal	5.4735740	['Sh', 'Tkl%
Christian N	5.8279793	['Lost_inv'
Declan Ric	5.3544926	['Att 3rd',
Eberechi E	5.2154466	['Sh', 'Prg'
Gabriel Ma	5.3089464	['Blocks', '
Gabriel Ma	5.0475571	['PrgR', 'T

Recommendation of players insights using Clustering

- Based on weakness, prompt GPT for training recommendation

name	pred_rating	weakness
Ben White	5.4046547	['SoTA', 'T
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Declan Ric	5.3544926	['Att 3rd',
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Gabriel Ma	5.3089464	['Blocks', '
Gabriel Ma	5.0475571	['PrgR', 'T

name	training_recommendations
Ben White	### Training Recommendations 1. Improve SoTA (Successful Tackles and Aerials): - Recommendation: Conduct targeted drills focusing on tackling and aerial duels. Set up 1v1 scenarios where Ben White is required to engage in both ground tackles and aerial challenges against a variety of opponents. Use video analysis to review his positioning and timing during these duels. - Reasoning: As a defender, winning tackles and aerial duels is crucial for regaining possession and preventing opposition attacks. Improving these skills will help Ben White disrupt the opposition's play more effectively, reducing the pressure on the defense and allowing the team to transition into attack more smoothly. 2. Enhance Tkl+Int (Tackles and Interceptions): - Recommendation: Implement situational awareness exercises that simulate game scenarios where Ben White must anticipate and intercept passes. Use small-sided games to emphasize reading the opponent's play and positioning himself to cut off passing lanes. - Reasoning: By improving his ability to intercept passes, Ben White can break up opposition plays before they develop into dangerous attacks. This will not only strengthen the team's defensive solidity but also create opportunities for quick counter-attacks, enhancing the team's goal-scoring potential. 3. Strengthen Def 3rd (Defensive Third Performance): - Recommendation: Focus on defensive organization and communication drills within the defensive third. Practice scenarios where Ben White must coordinate with his fellow defenders to maintain a compact shape and effectively mark opponents. Use video sessions to analyze successful defensive third strategies from top teams. - Reasoning: A strong performance in the defensive third is essential for preventing goals. By improving his understanding and execution of defensive strategies, Ben White can help maintain a solid defensive line, reducing the likelihood of conceding goals and allowing the team to focus on building attacks from a secure base.

Dream 11 Prediction using Clustering

- Calculated total team ratings for each formation
- Identified the formation with the highest overall rating

```
"formation": "4-3-1-2",
"team": [
  {
    "pos": "FW",
    "name": "Odsonne Édouard",
    "pred_rating": 5.722993585008704
  },
  {
    "pos": "FW",
    "name": "Sasa Kalajdzic",
    "pred_rating": 5.598138859137577
  },
  {
    "pos": "AM",
    "name": "Jadon Sancho",
    "pred_rating": 5.794616597315553
  },
  {
    "pos": "CM",
    "name": "Christian Nørgaard",
    "pred_rating": 5.826576767552558
  },
  {
    "pos": "DM",
    "name": "Mateo Kovačić",
    "pred_rating": 6.074856470991502
  },
  {
    "pos": "CM",
    "name": "Rayan Cherki",
    "pred_rating": 5.492304994267768
  }
]
```

